

AI-Powered Predictive Maintenance for Industrial Equipment

Risk Assessment & Mitigation Plan

Identifying Risks and Solutions

Team Members

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1. Risk Matrix Overview

The following risk assessment covers all major project risks across data, model, technical, deployment, team, and schedule dimensions. Each risk is scored by impact and likelihood.

ID	Category	Risk Description	Impact	Likelihood	Score	Mitigation Strategy
R1	Data Quality	Insufficient or noisy sensor data in CMAPSS/MIMII datasets	High	Medium	High	Apply robust preprocessing (imputation, outlier removal, resampling); use Librosa for audio denoising on MIMII data.
R2	Data Quality	Class imbalance in failure/anomaly labels	High	High	Critical	Use SMOTE or weighted loss functions; evaluate using F1/ROC-AUC rather than accuracy.
R3	Model Performance	LSTM model underfitting on RUL prediction	High	Medium	High	Tune sequence length, hidden units, and dropout; experiment with Bi-LSTM or Transformer variants.
R4	Model Performance	Autoencoder fails to distinguish normal vs. anomalous audio	High	Medium	High	Adjust reconstruction error threshold; experiment with variational autoencoder (VAE).
R5	Technical	Overfitting due to limited labeled data	High	Medium	High	Apply regularization (dropout, L2), data augmentation, and time-series cross-validation.

ID	Category	Risk Description	Impact	Likelihood	Score	Mitigation Strategy
R6	Technical	MLflow tracking instability or version conflicts	Medium	Low	Medium	Pin library versions; maintain a virtual environment; document all dependencies in requirements.txt.
R7	Deployment	FastAPI deployment failure or latency issues	High	Low	Medium	Conduct load testing; implement async endpoints; containerize with Docker for reproducibility.
R8	Deployment	Model drift post-deployment	High	Medium	High	Set up automated monitoring with drift detection; trigger retraining pipeline on performance degradation.
R9	Team	Uneven workload distribution causing bottlenecks	Medium	Medium	Medium	Weekly sync meetings; Alaa serves as technical lead to unblock dependencies; tasks reassigned proactively.
R10	Schedule	Delays in model training due to computational constraints	Medium	Medium	Medium	Use Google Colab Pro or Azure ML compute; prioritize GPU resources for LSTM and Autoencoder training.

2. Risk Ownership

ID	Owner(s)	Monitoring Approach
R1	Alaa, Taghreed, Nada	Review at weekly team meeting; escalate if score worsens. Tracked via MLflow metrics and model evaluation reports.
R2	Alaa, Nada	Review at weekly team meeting; escalate if score worsens. Tracked via MLflow metrics and model evaluation reports.
R3	Rowan, Alaa	Review at weekly team meeting; escalate if score worsens. Tracked via MLflow metrics and model evaluation reports.
R4	Alaa	Review at weekly team meeting; escalate if score worsens. Tracked via MLflow metrics and model evaluation reports.
R5	Alaa, Rowan, Nada, Taghreed	Review at weekly team meeting; escalate if score worsens. Tracked via MLflow metrics and model evaluation reports.
R6	Alaa, Nada	Review at weekly team meeting; escalate if score worsens. Tracked via MLflow metrics and model evaluation reports.
R7	Alaa	Review at weekly team meeting; escalate if score worsens. Tracked via MLflow metrics and model evaluation reports.
R8	Nada	Review at weekly team meeting; escalate if score worsens. Tracked via MLflow metrics and model evaluation reports.
R9	All	Review at weekly team meeting; escalate if score worsens. Tracked via MLflow metrics and model evaluation reports.
R10	Alaa, Rowan	Review at weekly team meeting; escalate if score worsens. Tracked via MLflow metrics and model evaluation reports.

3. Risk Monitoring Plan

Risks will be reviewed at weekly team sync meetings. The following triggers will prompt immediate mitigation action:

- Model evaluation metrics (RMSE, R2, ROC-AUC) fall below agreed thresholds
- Training runs exceed allotted compute time budget by more than 20%
- A team member is blocked for more than 2 business days
- MLflow logs show unexpected spikes in loss or drift indicators
- Deployment health check returns error rates above 1%